

Stress Engineering - Calculating Comparative stress and Tensile Strength

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Introduction

This project investigates the relationship between the **thickness** and **weight** of various materials, primarily paper-based, including single and layered A4 sheets as well as origami paper. The goal is to understand how increasing the thickness of a material affects its weight, and whether a predictable mathematical relationship exists between these two physical properties.

By collecting data on different material types and measuring their respective thicknesses and weights, we aim to analyse the correlation between these variables. Visualizations such as scatter plots, bar graphs, and regression models (both linear and polynomial) are used to reveal trends and patterns in the data.

This analysis not only helps in understanding the physical behaviour of paper materials under layering but also provides useful insights into material selection for design, engineering, or packaging applications where weight and thickness are critical factors.

Experiment

Aim:

To investigate the relationship between the thickness and weight of various paper materials, and to explore how these factors influence the stress-bearing capacity and tensile strength of the materials underload.

Apparatus required: Paper of different thickness and material, slotted weights, string, paper puncher, Transparent tape, paper cup

Formula and explanation:

In this experiment, we analysed the mechanical behaviour of different paper materials by calculating two key values: Comparative Stress and Tensile Strength. These values help us understand how each material responds to applied force, based on its thickness.

$$1. \text{Comparative Stress} = \frac{\text{Weight(g)}}{\text{Thickness(cm)}} \quad \text{Unit(g/cm)}$$

This formula gives a relative measure of how much mass (or load) a material can support per unit thickness

2. Tensile Strength

$$\text{Tensile Strength} = \frac{\text{Force(N)}}{\text{Cross-Sectional Area(cm}^2\text{)}} \quad \text{Unit(N/cm}^2\text{)}$$

This is a standard measure of the strength of a material, showing the maximum force, it can withstand per unit area before breaking.

Theory:

This experiment investigates the relationship between the thickness and weight of different types of paper materials, with a focus on understanding their mechanical behavior under load. Materials such as A4 sheets, origami paper, sticky notes, newspapers, tissue paper, notebook paper, sandpaper, and thicker variants of A4 sheets were tested. In physics, stress is defined as force per unit area ($\sigma = F/A$), and it is a key factor in determining how materials deform or fail under applied loads. While actual stress calculations require the full cross-sectional area, for the purposes of this investigation, comparative stress was calculated as weight divided by thickness, assuming a constant width across all samples. This allows us to compare how different paper types respond to increasing mass.

Procedure:

1. Use the paper hole punch to cut a hole about $\frac{1}{4}$ inch (0.63cm) from one end of the paper strip.
2. Use the pencil to make two holes under the rim of the cup-one on either side.
3. Tie one end of the string through one hole in the cup.
4. Thread the string through the hole in the paper strip, then tie the string in the second hole of the cup.
5. Secure the free end of the paper strip to the edge of a table with tape.
6. Allow the paper and cup to hang freely.
7. Add a few washers at a time to the cup until the string cuts through the paper.

Gallery:



Tabular column:

S. No	Material	Thickness (Measured Using a Vernier Caliper) (cm)			Weight Held (grams)	Comparative Stress (g/cm)	Tensile Strength (N/cm ²)
		Main Scale	Vernier Scale Reading	Final reading			
1	A4 sheet	0	9 x 0.01 = 0.09	0.09	300	3333.33	326.67
2	A4 sheet x 2	0.1	7 x 0.01 = 0.07	0.17	700	4117.65	403.53
3	A4 sheet x 3	0.3	3 x 0.01 = 0.03	0.35	1250	3571.43	350.00
4	A4 sheet x 4	0.5	7 x 0.01 = 0.07	0.57	2900	5087.72	498.60
5	Origami paper	0	2 x 0.01 = 0.02	0.02	240	12000.00	1176.00
6	Sticky note	0	2 x 0.01 = 0.02	0.02	300	15000.00	1470.00
7	Newspaper	0	1 x 0.01 = 0.01	0.01	150	15000.00	1470.00
8	Sandpaper	0	7 x 0.01 = 0.07	0.07	400	5714.29	560.00

9	Thicker A4 sheet	0	$4 \times 0.01 = 0.04$	0.04	450	11250.00	1102.50
10	Tissue paper	0	$1 \times 0.01 = 0.01$	0.01	70	7000.00	686.00
11	Notebook	0	$7 \times 0.01 = 0.07$	0.07	120	1714.29	168.00

Calculation:

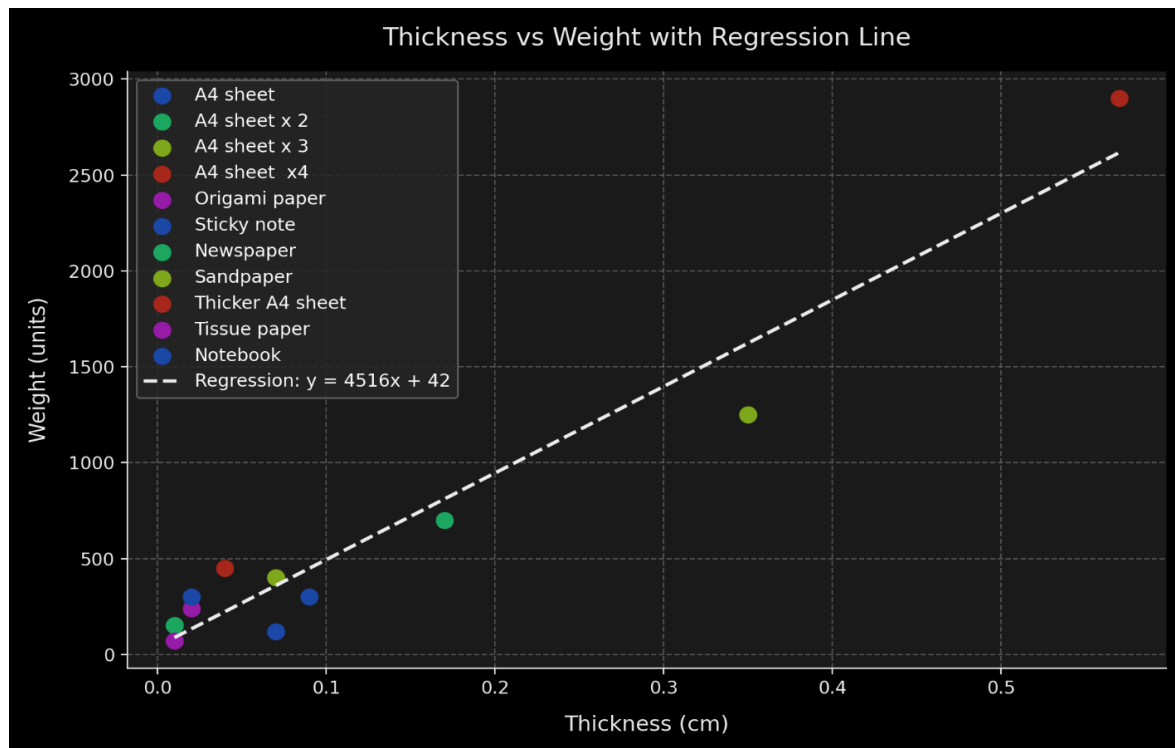
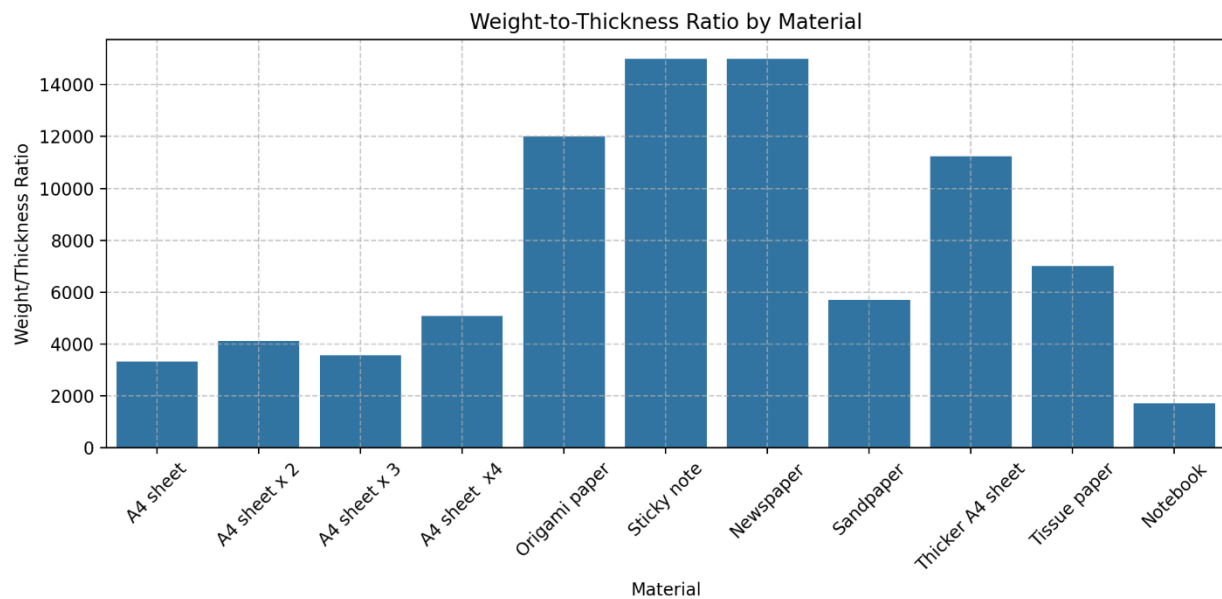
$$\text{Comparative Stress} = \frac{\text{Weight}}{\text{Thickness}} \quad (\text{g/cm})$$

Tensile Strength:

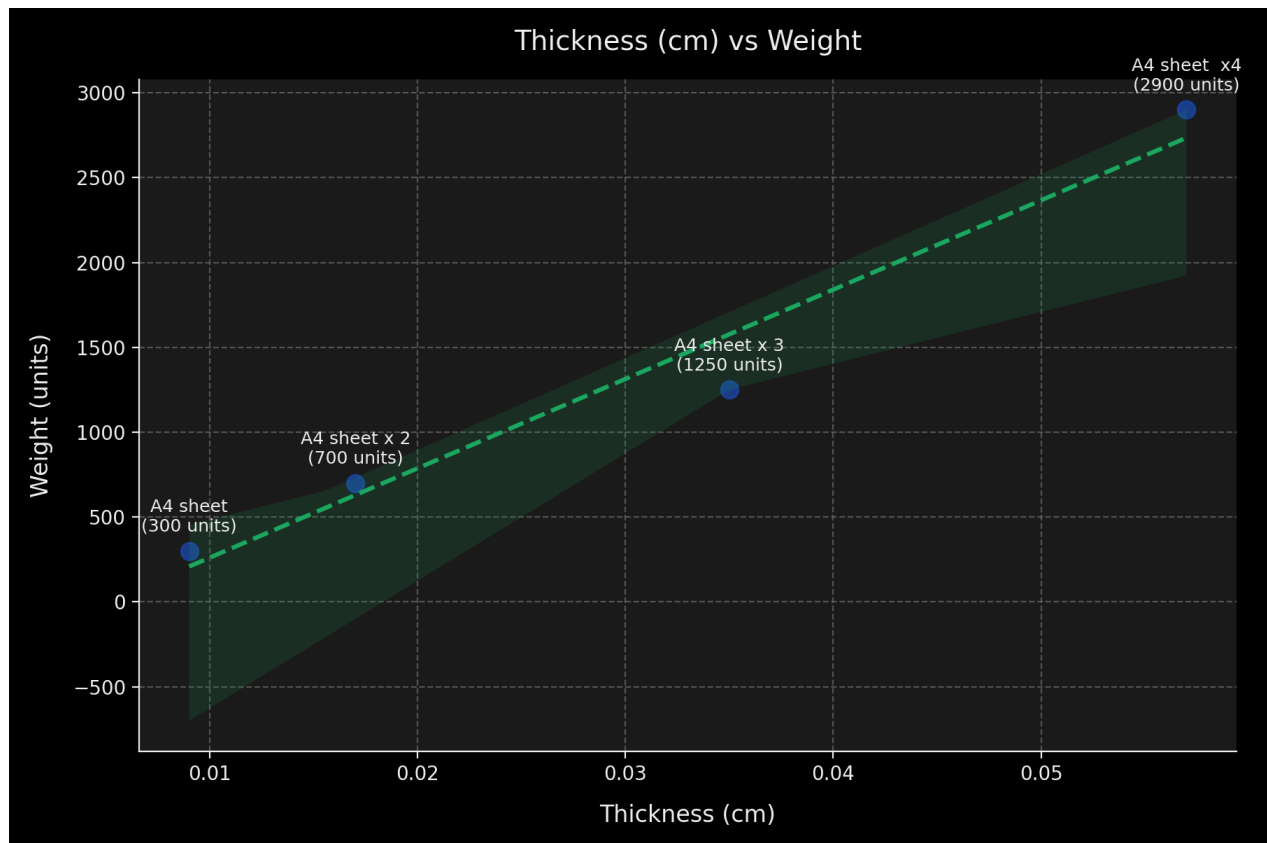
Tensile Strength = Force/ Cross-sectional area

The unit of tensile strength is N/cm^2

Graph:



A Regression line also known as the line of best fit, is a straight line that best represents the relationship between two variables in a dataset.



Result:

Key findings:

1. There's a very strong positive correlation (0.97) between thickness and weight
2. As the number of A4 sheets increases, both thickness and weight increase in a nearly linear fashion
3. Different materials have vastly different weight-to-thickness ratios - origami paper, sticky notes, and newspaper are much more efficient in terms of weight-to-thickness ratio compared to A4 sheets

Why Thinner Materials May Show Higher Stress:

1. Material Composition & Internal Structure

- Materials like newspaper and sticky notes are made from densely compressed pulp with longer cellulose fibres, making them tougher and more cohesive, even if thin.
- Origami paper is often treated or coated, giving it more tensile strength and rigidity than regular paper.

So even with less thickness, they resist deformation better and spread force more evenly.

It's not just about how thick a material is, but how it behaves underload its composition, structure, and how it transfers force make a huge difference.

Precautions and errors:

- Opt for lower weight loads to ensure measurements are more accurate.
- Make sure the paper is properly anchored to the table.
- Make sure the weights are placed gently onto the chosen cup to ensure more accurate measurements.

Conclusion:

From the experiment, it is evident that there is a strong relationship between the thickness and weight of paper-based materials. As the thickness of the material increases, its weight generally increases as well, indicating a positive correlation. The comparative stress analysis revealed that materials like sticky notes, newspapers, and origami paper exhibited high stress values, suggesting a greater weight per unit of thickness. This implies that these materials are either denser or more tightly packed than others like tissue paper or standard A4 sheets.

By using a comparative stress calculation ($\text{Weight} \div \text{Thickness}$), we were able to draw meaningful comparisons between different materials despite not measuring the full cross-sectional area.

Bibliography

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